

WASP-43b

R-0435

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250103 · created 2026-07-19T12:09:16+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460678.9776 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.022
Transit depth (Rp/Rs)^2	3.33 +/- 0.79 [%]
Orbital Inclination (inc)	80.03 +/- 2.16
Airmass coefficient 1 (a1)	0.879 +/- 0.011
Airmass coefficient 2 (a2)	0.09 +/- 0.13
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#3 - [376, 36]
Optimal Aperture	3.31
Optimal Annulus	5.86
Transit Duration (day)	0.04 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.022 vs 0.1594 (deviation 1.01σ)
Duration fit vs published	0.04 d vs 0.0512 d (deviation 1.01σ)
Mid-time O–C	6.6 min
Depth SNR	8.2
Residual scatter	1.29 %

Light curve

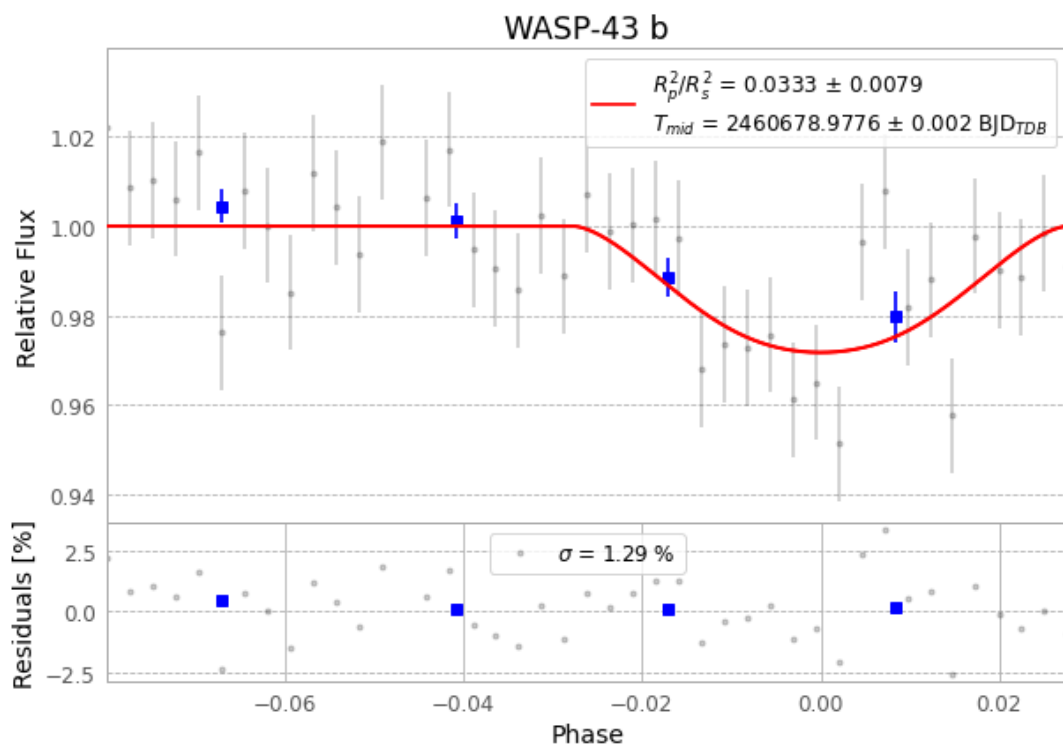


Figure 1 — FinalLightCurve_WASP-43 b_2025-01-03.png (SHA-256 47f73973fcf6a5fd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [453, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eed7dbbad6299486...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

43 FITS frames (28 MB), WASP-43250103094815.FITS → WASP-43250103115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250103.log (1b7ed31ad4da...), FinalLightCurve_WASP-43 b_2025-01-03.png (47f73973fcf6...), FinalLightCurve_WASP-43 b_2025-01-03.csv (1166d4134952...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 12:09Z.